

Exploring minimality in grammar change

Birgit Alber, Eirini Apostolopoulou & Nick Kalivoda

Apart from imperfect learning or covert reanalysis of structures, language change may be driven by learners' biases to override evidence to which they have been exposed (Garrett & Johnson 2013; Garrett 2014; see Kiparsky 2015 for an overview). Following these approaches, we propose that one bias determining how diachronic evolution advances is to keep change minimal in grammatical terms. Working in the framework of O(ptimality)T(heory)-based property theory (Alber & Prince 2015, in prep.), we define minimal grammatical change as a single resetting of the defining ranking conditions (properties) of a typological system (Alber 2015; Alber & Meneguzzo 2016; DelBusso 2018; Apostolopoulou 2022). We claim that phonological change does not necessarily imply minimal disruption of surface features, but can involve significant change in surface structure via the minimal resetting of grammatical parameters.

We illustrate our proposal via two case studies: (1) the variation of metrical parameters of word stress in the world's languages, e.g., foot type, parsing density, alignment, etc. (see Alber & Prince 2017, in prep); (2) long-distance metathesis of liquids in Romance, e.g., in terms of the syllabic position targeted by metathesis, the possible landing sites, and the distance covered by the migrating segment (see Apostolopoulou 2022). Both case studies draw on change affecting prosodic structure, which is generally rarely explored in historical linguistics (for an exception see Lahiri 2015), yet may offer clear evidence in favor of our claim that at least some minimal language change is equivalent to a change in grammar and not only on the surface.

We compare our property-based approach with another OT-based grammatical theory of change, that of *border point pairs* (Merchant 2019). In this theory, a minimal change involves transition between two grammars via minimal reranking of two constraints that are adjacent in a total order, e.g., $P \gg \underline{X} \gg Y \gg Q \rightarrow P \gg \underline{Y} \gg X \gg Q$ ($\underline{X} \gg Y \rightarrow Y \gg X$ being the border point). The predictions of the Border Points approach largely coincide with those of Property Theory (e.g., attested changes in stress in closely related languages spoken in Arnhem Land, Australia). However, we present a case of a change in foot type property observed in two Panoan languages, Saynáwa and Huariapano, which are *not* adjacent in terms of border point pairs, even though they differ only on *one* property value.

Finally, in light of the two typological systems employed in our analysis, we discuss what changes are predicted to occur and, in turn, what diachronic paths can be reconstructed to describe the transition between different grammars.

Selected references

- Alber, B. & A. Prince. In prep. The structure of OT typologies. ROA-1381.
Alber, B. & A. Prince. 2017. The Book of nGX. ROA-1312.
Apostolopoulou, E. 2022. Typological variation in language contact: A phonological analysis of Italiot Greek. PhD thesis, UniVR & UiT.
Kiparsky, P. 2015. New perspectives in historical linguistics. In C. Bowerman & B. Evans (eds.) *The Routledge Handbook of Historical Linguistics*, 64–102. Routledge.
Merchant, N. 2019. Distance from the Edge: Using Border Points to Construct Topologies. Paper presented at *OCP16*.